

Smart IoT-Based Vehicle Security & Driver Monitoring System

An Integrated Real-Time Platform for Intrusion Detection, Face Authentication & Drowsiness Monitoring

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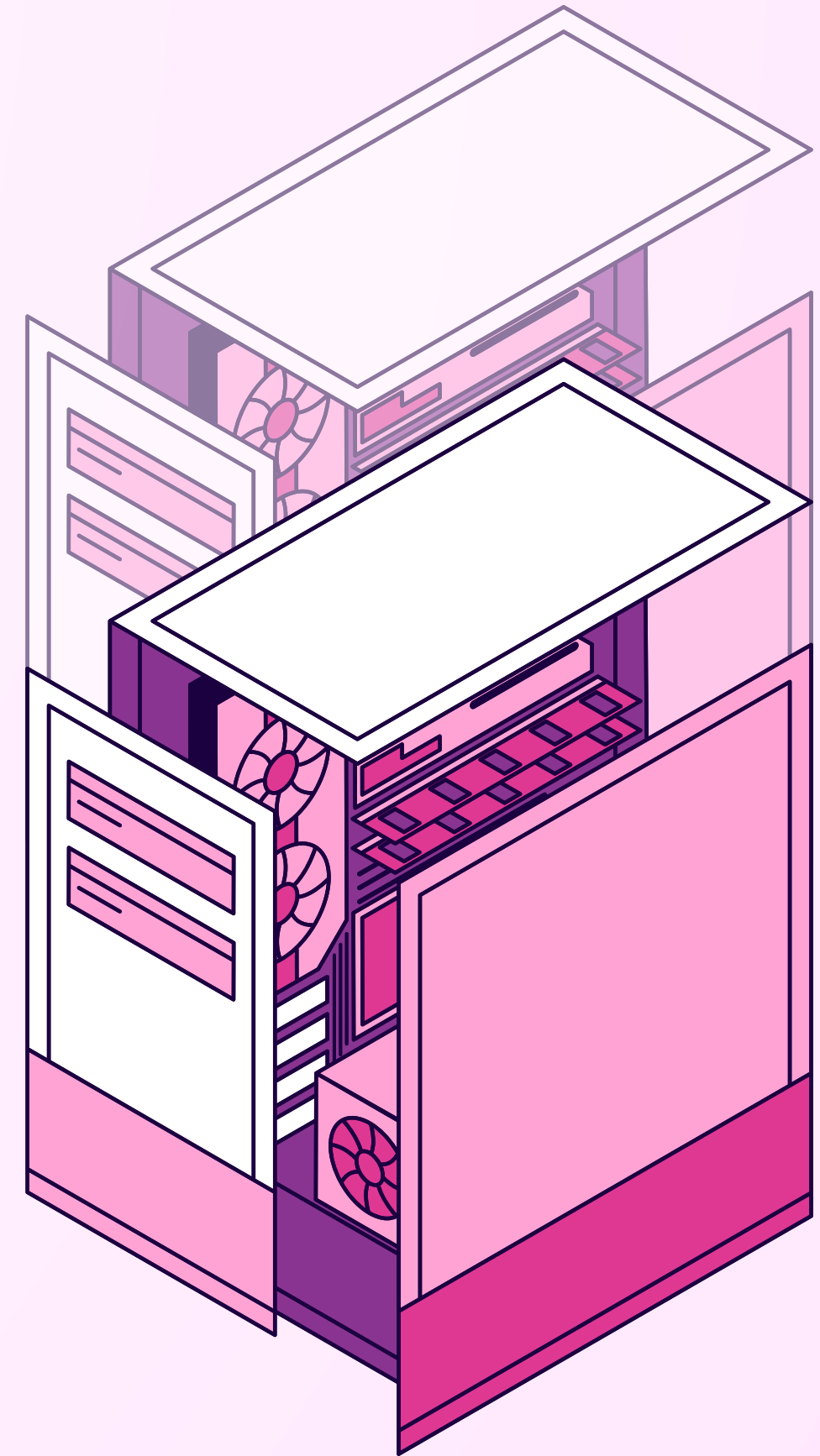
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Why do we need this system

- Vehicle theft & vandalism are rising globally
 - Traditional alarms trigger on false positives — no intelligence layer
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- Driver fatigue causes thousands of road accidents annually
 - Existing systems address security OR driver safety — never both
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- Cloud-only inference creates dangerous latency for real-time alerts



What We Set Out to Build

Multi-modal intrusion
detection (vibration + sound
+ pressure)

Face recognition-based
access control

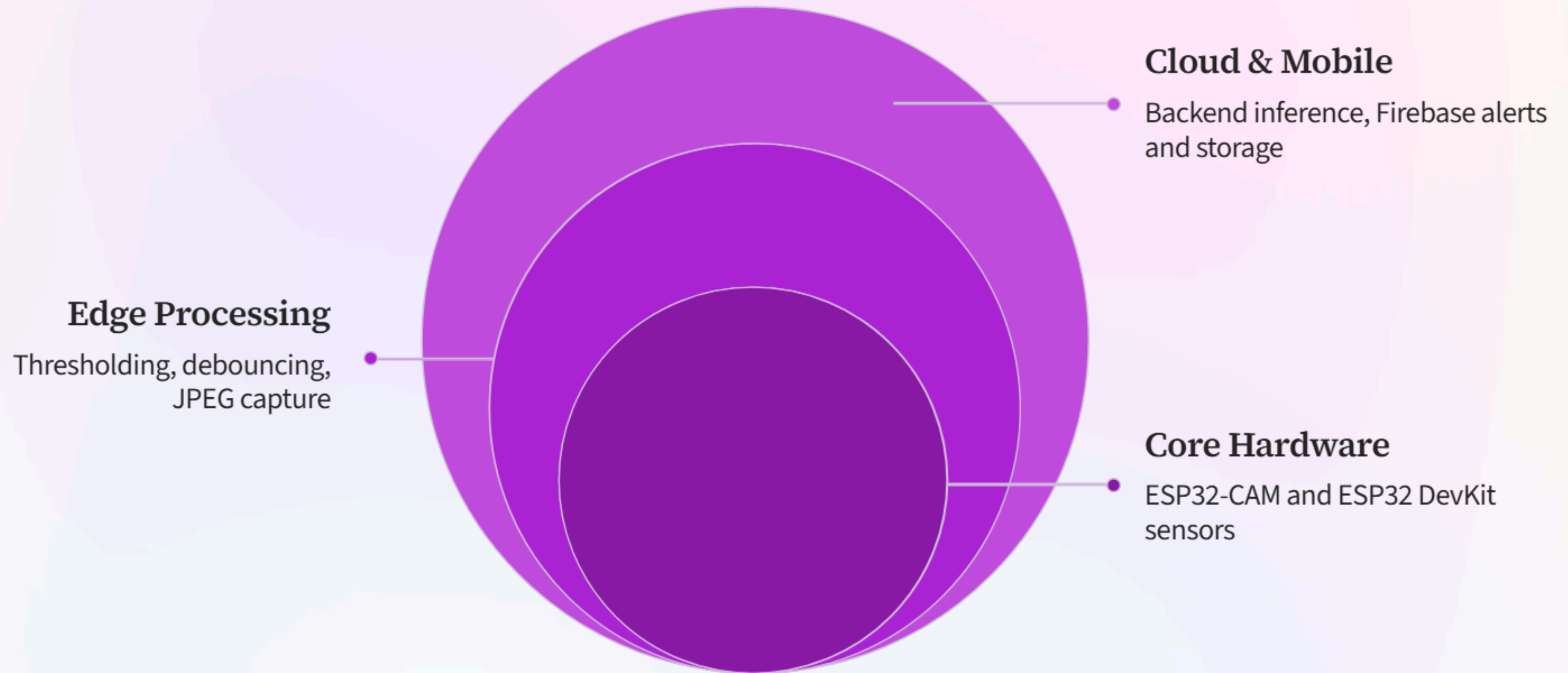
Real-time driver drowsiness
detection via Eye Aspect Ratio
(EAR)

Cloud integration for alert
logging + mobile push
notifications

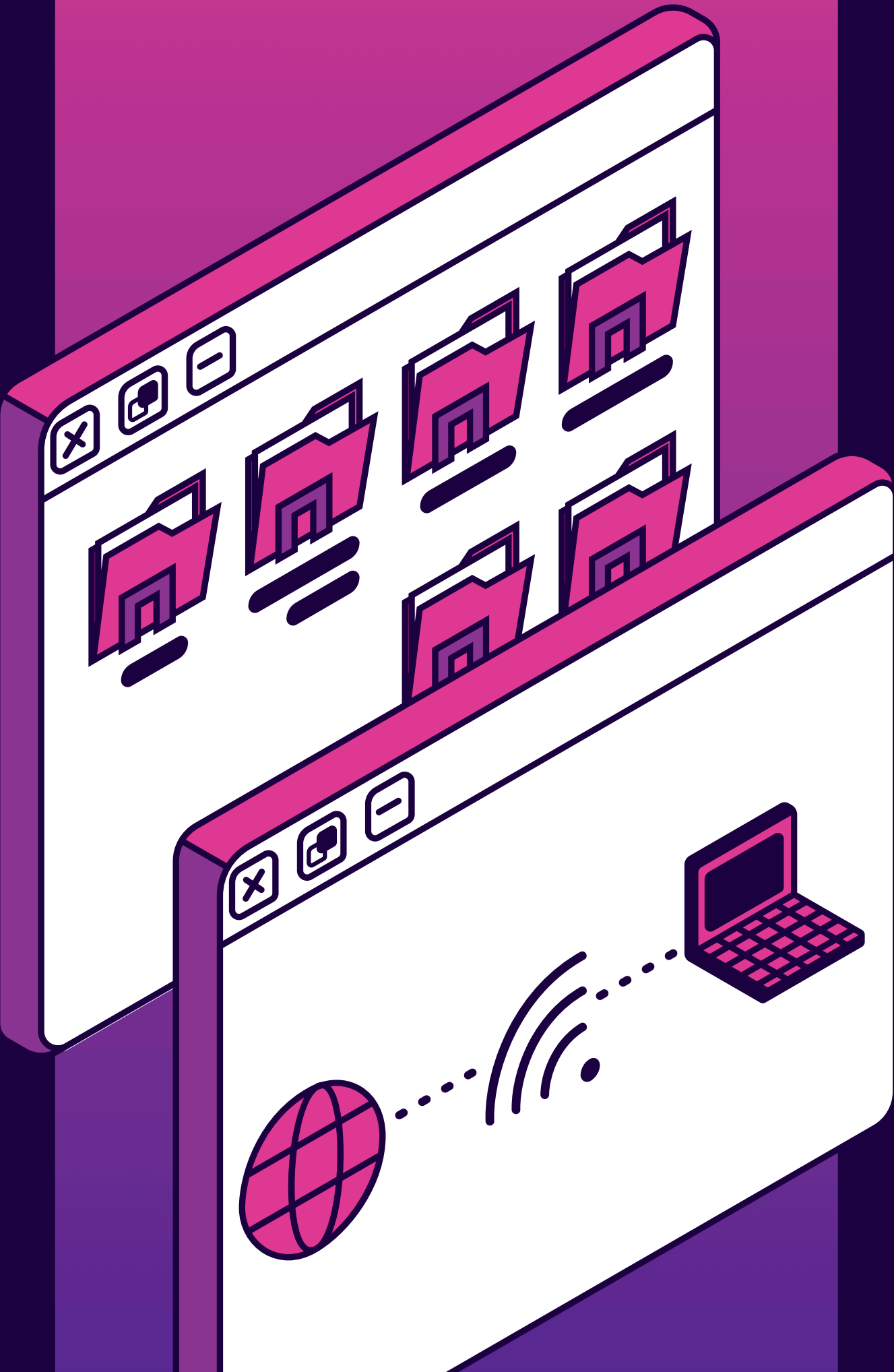
Cost-effective prototype
deployable in any personal
vehicle

TARGET COST: Rs. 1,500–2,000 (vs. expensive commercial alternatives)

Four-Layer System Architecture



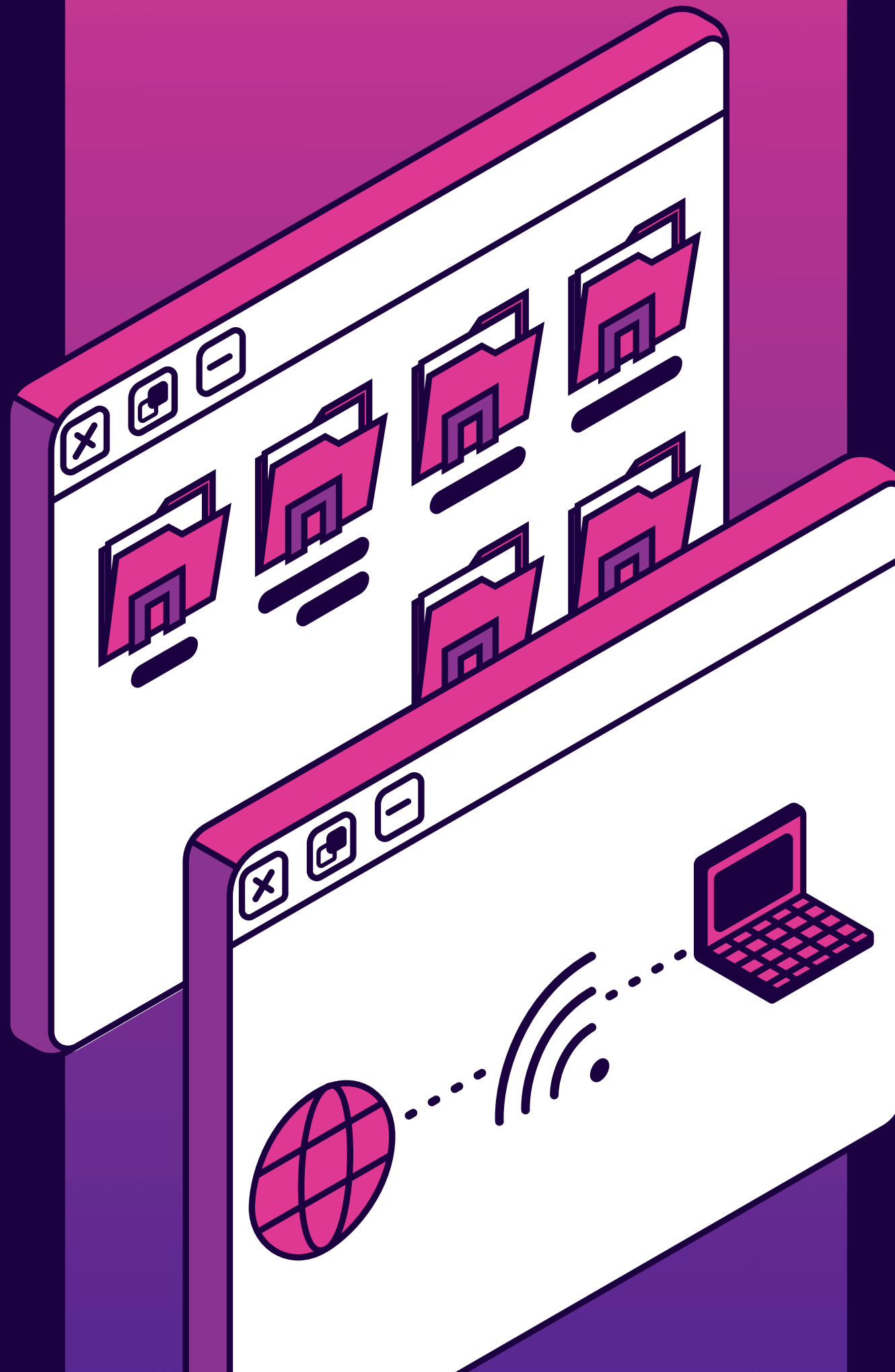
The dual-microcontroller architecture separates camera and sensor responsibilities — improving stability and enabling parallel operation across all four layers from



Hardware & Software Stack

Key Hardware Components

Component	Model	Purpose
Camera + Wi-Fi	ESP32-CAM	Image capture
Sensor MCU	ESP32 DevKit	Sensor aggregation
Vibration Sensor	SW-420	Impact detection
Sound Sensor	KY-037	Glass break
Load Cell + ADC	50kg + HX711	Roof pressure
Power Supply	3S Li-ion + Buck	5V / 3.3V rail

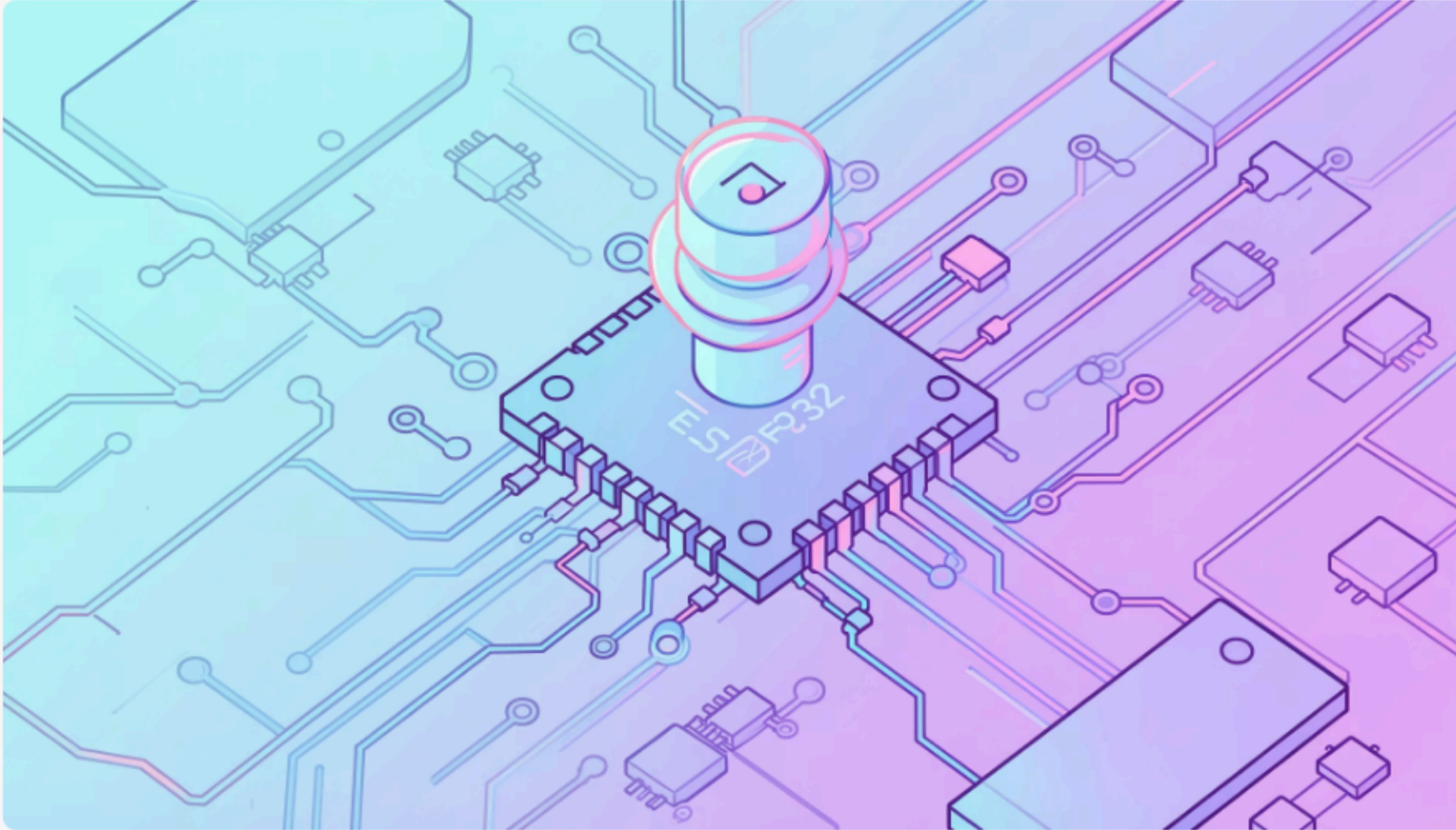


Hardware & Software Stack

Technology Stack

1. Embedded: Arduino C++ on ESP32 — sensor polling, UART, HTTP POST
2. Backend: Python Flask — face_recognition (dlib), MediaPipe + OpenCV for EAR
3. Cloud: Firebase Realtime DB, Cloud Storage, FCM push notifications
4. Frontend: Next.js + TypeScript — alert dashboard, auth status, drowsiness panel
5. Containerization: Docker for portable, scalable backend deployment

Module 1: Multi-Modal Intrusion Detection



How It Works

ESP32 polls SW-420, KY-037, and HX711 every **50ms**.
Sensor fusion with debounce + cooldown (8s)
significantly reduces false positives.

● CRITICAL

Sound + Vibration both confirmed

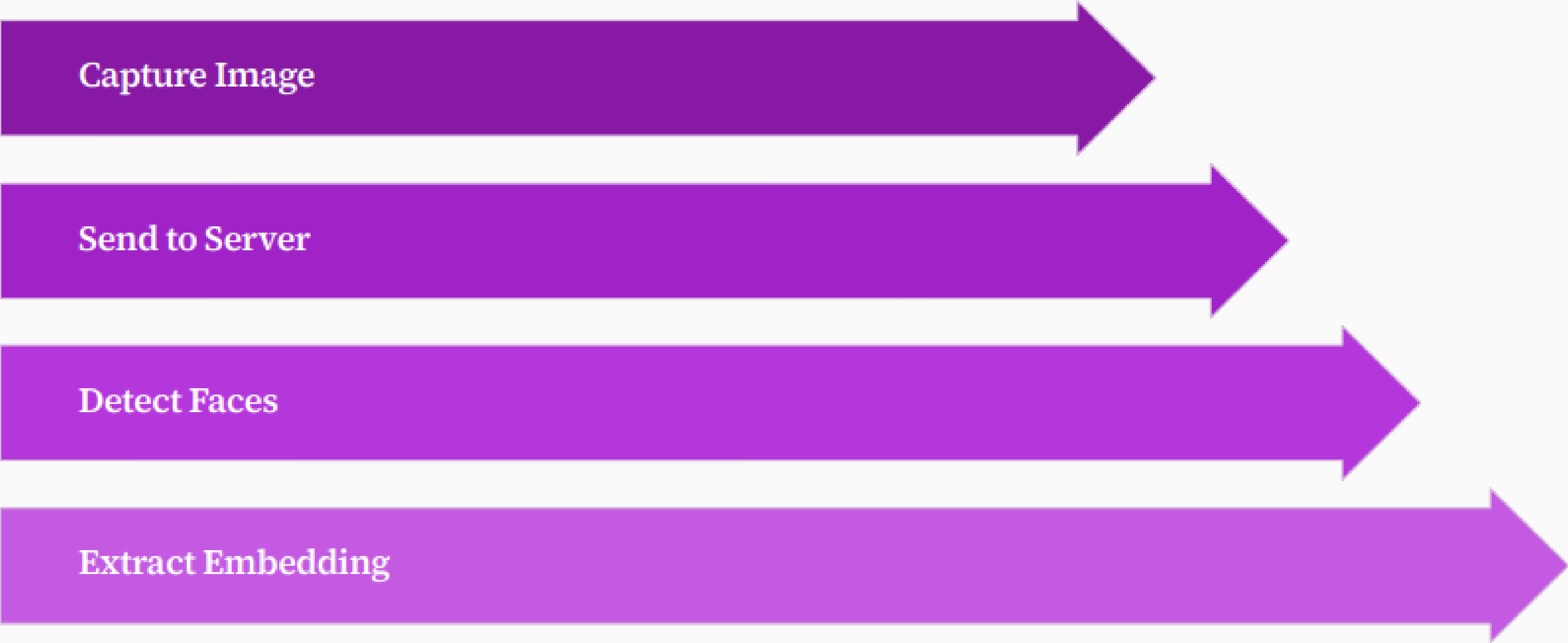
● HIGH

Pressure sensor + any secondary

● LOAD CELL

Weight > 3 kg triggers independently

Module 2: Biometric Face Authentication



Capture Image

Send to Server

Detect Faces

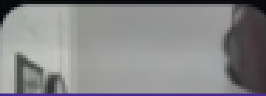
Extract Embedding

Biometric Face Authentication

Algorithm Highlights HOG-based face detection followed by 128-dimensional ResNet embeddings. Euclidean distance threshold of 0.45 determines authorization. Results logged to Firebase with FCM push notification.

Date & Time	Action
Apr 21, 2026 7:08:31	 Ignore  Inspect
Apr 21, 2026 0:59:40	 Ignore
Apr 21, 2026 0:56:49	 Ignore  Inspect

Unauthorized Access Attempts

Image	Description
	suspicious activity
	Faces detected: 0
	

Performance 99.38% accuracy on LFW benchmark (face_recognition / dlib) 93.3% system accuracy under mixed lighting Tested on 3 authorized users + 10 unauthorized individuals

< Alerts

View and manage all intrusion alerts

Total Alerts

12

Ignored Alerts

2

Unauthorized Attempts

9

Unauthorized Access Attempts

Image	Description	Date & Time	Actions	
	suspicious activity	Apr 21, 2026 17:08:31	Ignore	Inspect
	Faces detected: 0	Apr 21, 2026 10:59:40	Ignored	
	Faces detected: 0	Apr 21, 2026 10:56:49	Ignore	Inspect

Recent Drowsiness Alerts

Mild	↻ 24 km/h	15 Apr 2026, 05:25 PM
Mild	↻ 24 km/h	15 Apr 2026, 05:25 PM
Mild	↻ 21 km/h	21 Apr 2026, 07:39 PM

Dashboard

Overview of your vehicle security system

Total Alerts

6

-12% from last week



Daily Logs

5

+8% from last week



System Status

Active



Quick Access



Intrusion Detection

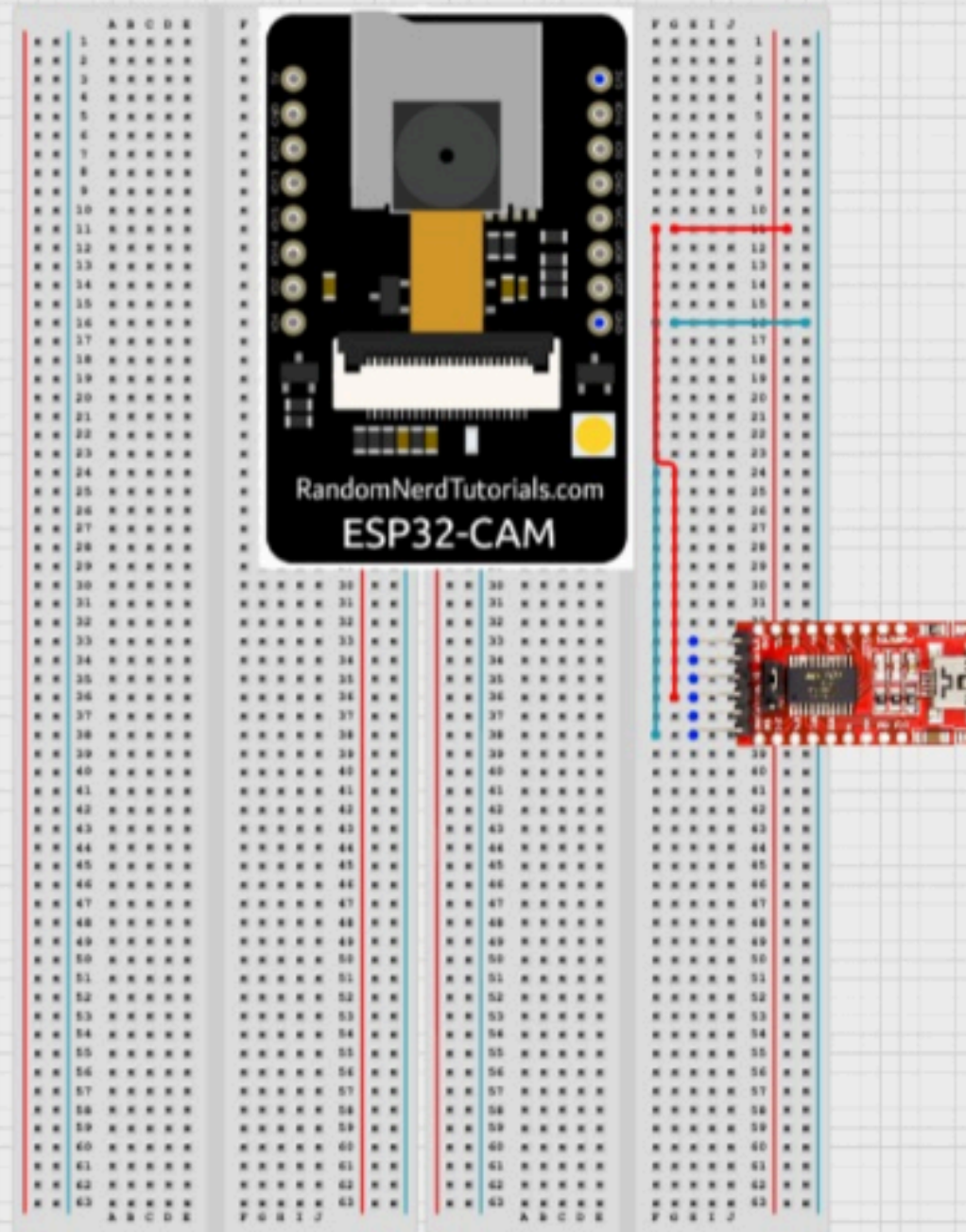
Monitor and manage security alerts, view daily access logs, perform live inspections, and configure relax mode for your vehicle.



Drowsiness Detection

Track driver alertness levels, view drowsiness alerts and analytics, and ensure safe driving conditions.

Module 3: Real-Time Driver Drowsiness Monitoring



EAR FORMULA:

$$\text{EAR} = (\|p2-p6\| + \|p3-p5\|) / (2 \times \|p1-p4\|)$$

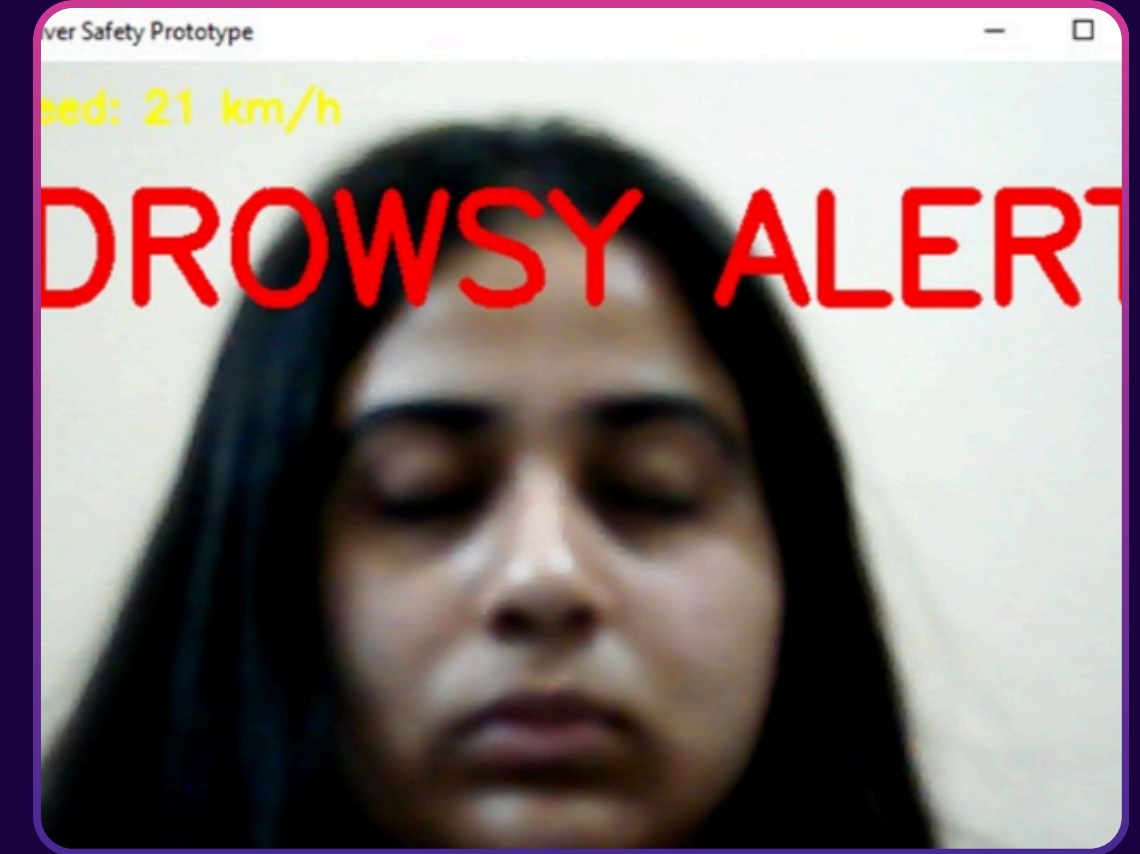
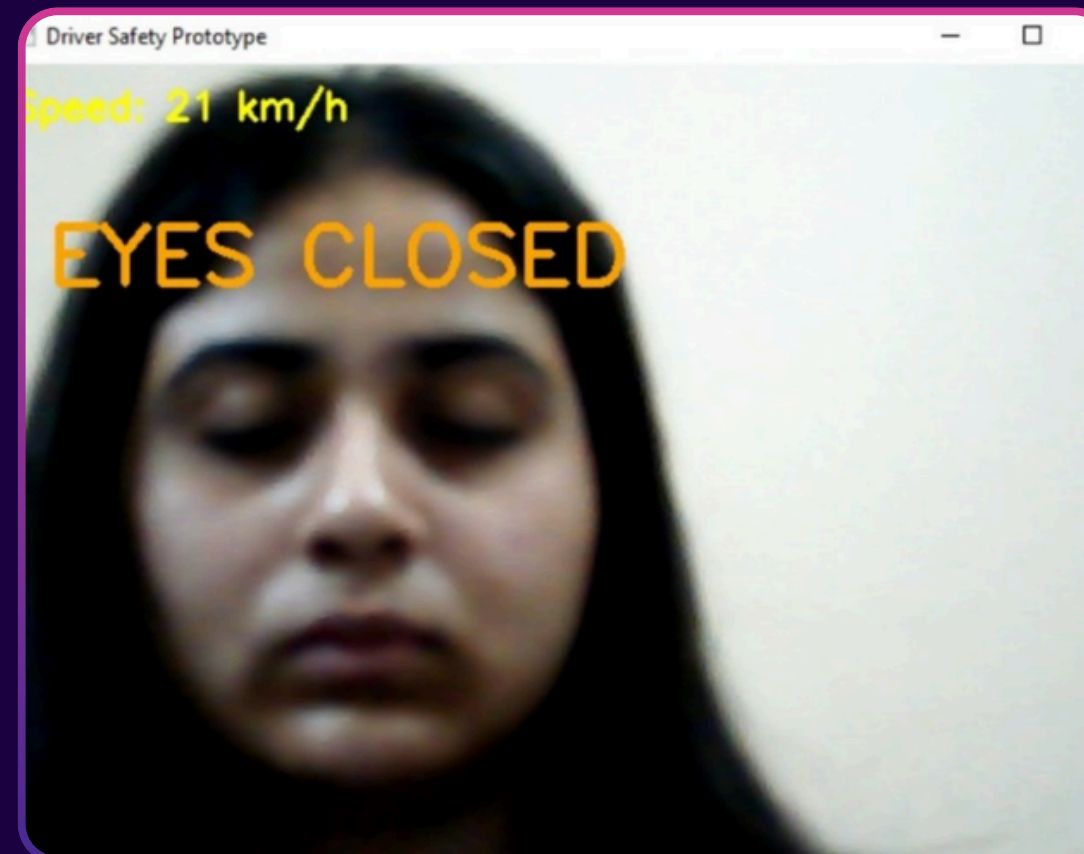
THRESHOLDS:

- $\text{EAR} < 0.22 \rightarrow$ Eyes detected as closed
- Eyes closed continuously > 2.5 seconds $\rightarrow$ DROWSINESS ALERT
- Alert triggers buzzer + Firebase push notification

SYSTEM ACCURACY: 90% across 30 test events

HOW IT WORKS:

- Activates only when vehicle speed exceeds 20 km/h
- Continuous MJPEG stream sampled at ~10 fps
- MediaPipe Face Mesh detects 468 3D facial landmarks per frame
- Eye Aspect Ratio (EAR) computed from 6 landmark points per eye



Hardware components & connections

Load Cell → Debounce Fix

Extend debounce to 5 consecutive readings

Face Recognition → IR LED

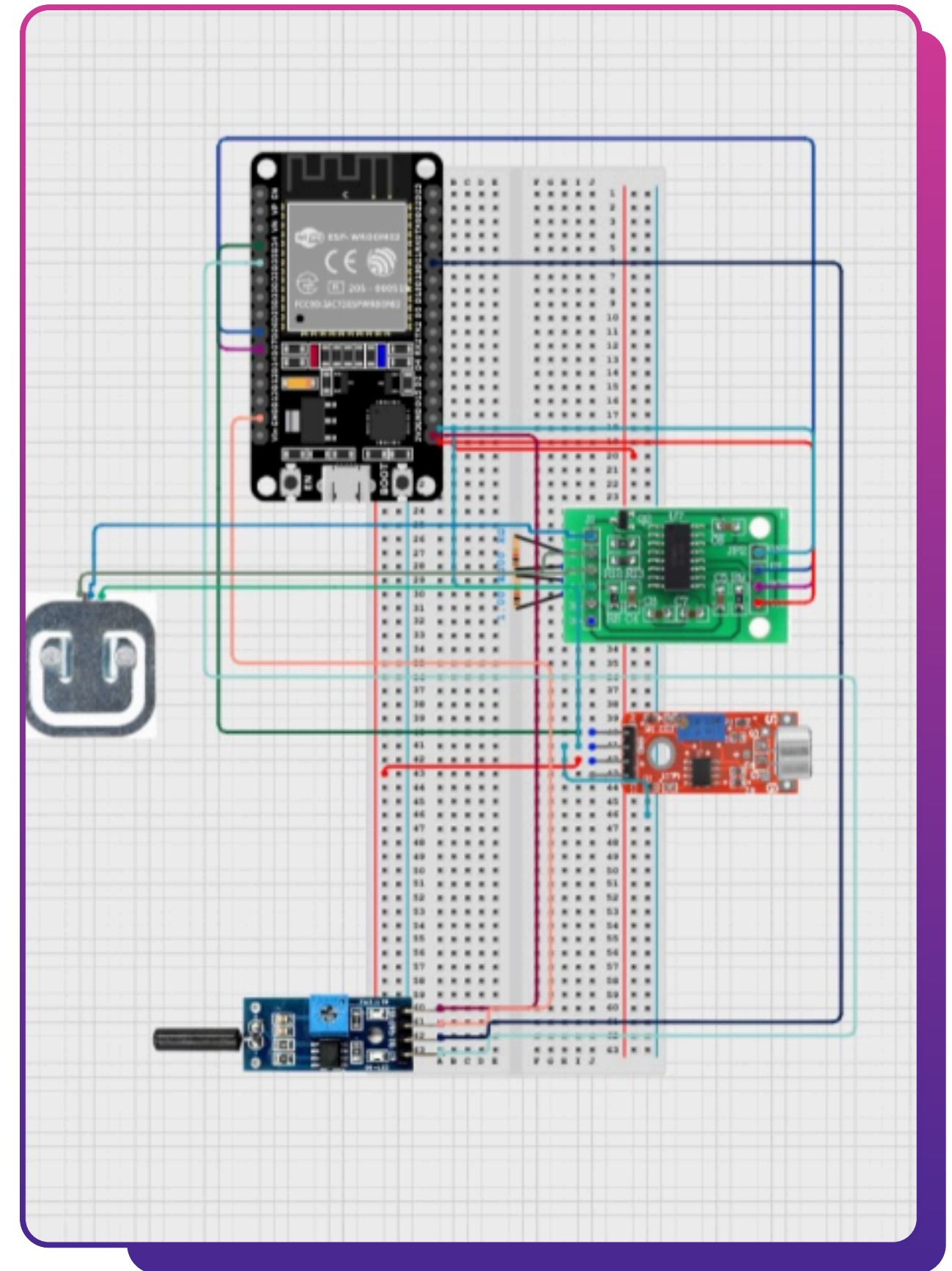
Add IR LED adjacent to camera for low-light accuracy

Sound Sensor → Mic Upgrade

Replace KY-037 with analog mic + frequency-domain analysis

Drowsiness → Head Pose Filter

Add head-pose estimation to handle turned-head cases



Advantages, Future Work & Conclusion

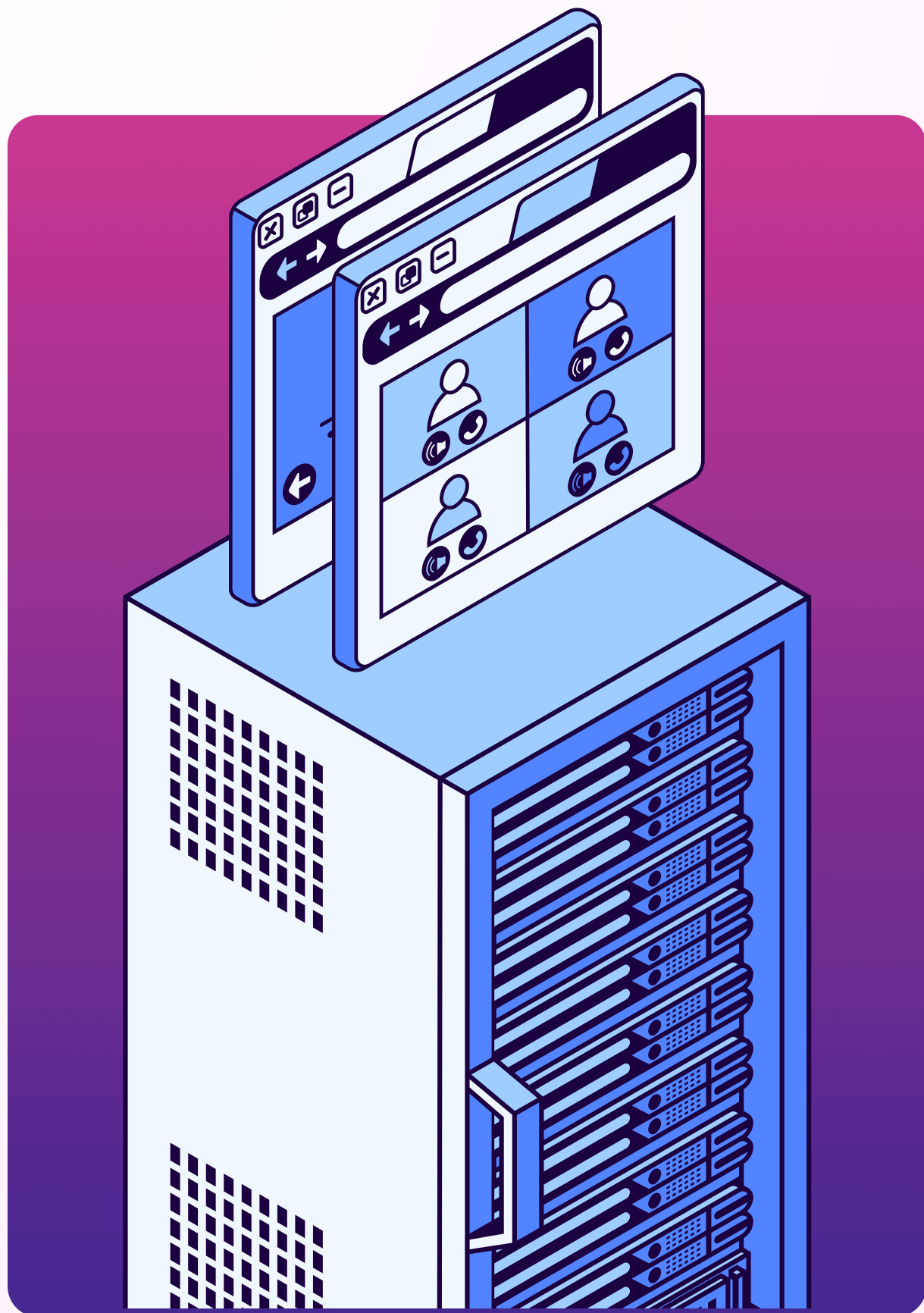
Competitive Edge

Prior systems (Kumar 2019, Sharma 2020, Lalitha 2021, Patil 2018) each address only one dimension — GSM-only, RFID-only, desktop-only EAR. Our platform uniquely integrates intrusion + face auth + drowsiness in one affordable system.



What's Next

- Raspberry Pi 4 — fully embedded backend
- On-device face recognition via TensorFlow Lite
- Battery Management System for Li-ion safety
- Enhanced mobile app with live camera feed
- GPS integration for real-time theft tracking



THANK YOU